

Self-Play Reinforcement Learning for the Pokémon TCG

One-Page Project Report

Objective

We developed a reinforcement learning agent for Pokémon Trading Card Game (PTCG) using self-play. The objective was to learn game-playing behaviour across a diverse collection of decks rather than depending on hand-written decision rules.

Approach

The game state was represented as a sequence of tokens covering the player's and opponent's cards across their decks, hands, active and bench Pokémon, stadiums, discard piles, and game statistics. Legal actions were also represented as tokens containing their source, target, type, and relevant attributes.

A 20M-parameter Transformer was used with policy and value heads. The policy selected the next action, while the value head estimated the current game state. The same model could act from either player's perspective during self-play.

Self-Play Training

The agent was trained using PPO with Generalized Advantage Estimation. A frozen teacher checkpoint was used for periodic head-to-head evaluation. To expose the model to diverse strategies, the training pool contained 4,941 decks collected from competition data and competitive decklists.

The self-play experiment consisted of approximately 50M game steps. Training-game outcomes were recorded and used to analyse performance across different deck archetypes. Efficient state encoding and engine-side optimisation were used to make large-scale self-play practical.

Evaluation and Final Stage

Checkpoints were evaluated against a fixed collection of strong decks and through competition evaluation. Matchup results were analysed to identify consistently strong archetypes. Alakazam and Dragapult+Dusknor were selected as the final deck archetypes, followed by targeted fine-tuning for the final agents.

Key Results

- 20M-parameter Transformer model.
- 4,941 decks represented in the training pool.
- Approximately 50M self-play game steps.
- Multiple deck archetypes evaluated through recorded game outcomes.
- Final competition standing: top 10.53%.

Conclusion

The project demonstrated a self-play RL approach for learning Pokémon TCG gameplay across a large and diverse deck pool. Token-based state and action representations allowed the Transformer to process complex game states, while PPO self-play provided the learning framework. Analysis of the resulting games helped identify suitable final deck strategies and guided targeted fine-tuning.